Plant Leaf Disease Detection Using AI Models

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1. INTRODUCTION

For this project, I worked on a plant leaf disease classification task using machine learning and deep learning techniques. I chose this dataset because plant diseases can affect crop quality and productivity, and early detection can help reduce damage. The main goal was to compare different models and determine which one performs best in classifying plant diseases from leaf images.

2. DATASET DESCRIPTION

The project used the PlantVillage dataset, which contains 20,638 images of plant leaves. The dataset includes 15 classes representing healthy leaves and different diseases affecting tomato, potato, and pepper plants. Before training, the images were resized and normalized to improve model performance. The dataset was then divided into training and testing sets using an 80/20 split.

3. MODELS USED

Three models were trained and evaluated in this project:

- Decision Tree (F1-score = 39.38%)
- Random Forest (F1-score = 65.11%)
- Convolutional Neural Network (CNN) (F1-score = 85.57%)

Among all models, the CNN achieved the highest performance and produced the most accurate predictions.

4. NEURAL NETWORK

The neural network was built using convolutional layers, max-pooling layers, and dense layers. These layers helped the model learn important visual features from plant leaf images. Dropout was also used to reduce overfitting and improve generalization. After training, the CNN achieved an accuracy of 85.63% on the test dataset, which was higher than the other models.

5. RESULTS & COMPARISON

The results showed that the CNN outperformed both the Decision Tree and Random Forest models. While the Random Forest produced acceptable results, it was not as effective as the CNN in handling image data. The Decision Tree had the lowest performance among the tested models. Since CNNs are specifically designed for image classification tasks, they were able to learn more useful patterns and features from the plant leaf images. Based on the evaluation metrics, the CNN was selected as the final model.

6. WHAT I LEARNED

This project helped me understand the complete machine learning workflow, including data preprocessing, model training, evaluation, and comparison. I also learned the importance of selecting a model that matches the type of data being used. In addition, I gained practical experience working with image datasets and building an AI application that can assist in plant disease detection.